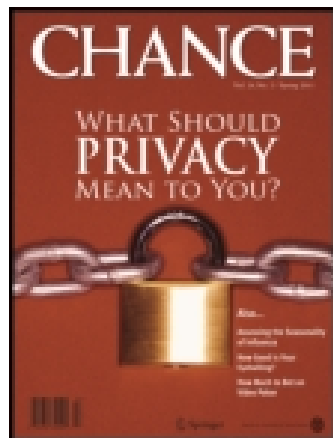


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A Statistician Reads the Sports Pages

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A STATISTICIAN READS THE SPORTS PAGES

**Scott M. Berry,
Column Editor**



My Triple Crown

In writing this column four times a year I find too many "small article" topics; topics that are interesting but not substantial enough for a whole column. This issue is an attempt to address three of these smaller, yet interesting, topics. It is my attempt at a ... *triple crown*! In the first leg of the triple crown I address the all-or-nothing strategy in hockey of pulling the goalie when you are down by a goal late in the game. Losing by two goals is no different from losing by one goal—and the chance to have six "skaters" against the opponent's five increases the chances of tying the game. If you pull the goalie too early you may give up an empty net goal, which essentially ends your chances of tying the game. Should the goalie be pulled, and if yes, when is the optimal time to do so?

In the second part of this article I present a neat little model for predicting the results of the NCAA basketball tournament. There have been several great articles on this subject—I do not dispute anything written in those; I just present a neat little model for the quality of the seeds and address the expected number of upsets. I found it very interesting that there were several Internet sites offering \$1,000,000 to anyone who could correctly predict all 63 games of the 2000 tournament ... how likely is it that someone will win?

In the last section of the article I try to address a nasty, yet important, question in sports and statistics. In my second *Chance* column (Vol. 12, No. 2, Spring 1999) I predicted the 1999 season home-run totals for McGwire and Sosa, as well as 25 other players. Well, the season is over, and I frequently get asked the very simple question, "Were you correct?" Well, of course, I did not get each player's season total exactly correct, but I do not think that is the question. In the final leg of my triple crown I try to interpret and answer this simple but nasty question.

First Leg: Pulling the Goalie

One of my stranger memories of playing goalie for 20 years in the frozen Minnesota winters occurred when I was being

"pulled." My team was down by one goal, late in the third period. The coach signaled me to go to the bench for a skater. This would allow our team a better chance to tie the game and at the same time greatly increase the chances of losing by two goals. As I was on my way to the bench the puck came to me. I—picture me in a ton of equipment with a stick made for stopping pucks not shooting them—controlled the puck and actually faked out an opponent and carried it across the red line (okay, so he thought I was going to dump it). As I was preparing to get rid of the puck, the whistle blew. Little did I know it was a penalty for the goalie to carry the puck past the red line! Well, we lost the game, but it was neat for me to tell people that I, as the goalie, got a penalty for touching the puck past the red line.

Each team in hockey can have a maximum of six players on the ice at a time. Typically, one is the goalie and the other five are "skaters." Teams are forced to play without a skater when they receive a penalty (though they never have less than four players on the ice at a time). When a team is down by one goal, late in the game, it is common to remove the goalie in exchange for a skater. A unique feature of ice hockey is that this can be done during the continuous action; in hockey jargon this is referred to as "on-the-fly." The increase in skaters creates a 6-on-5 advantage that increases the offensive potential, but it also leaves the goal unprotected. Is the increase in the probability of scoring a goal worth the risk of giving up an easy open-net goal, which would essentially end any chance to tie the game?

To address this question I make an assumption that the time until the next goal is scored for a team is an exponential distribution with a mean of λ . I do not address the empirical validity of this assumption in this article, but I have looked at goals scored within a game and the Poisson distribution provides a good fit. If goal scoring for a team follows a Poisson process then the time between goals will have an exponential distribution. Others have also used this assumption (Anderson-Cook and Thornton 1998; Berry, Reese, and Larkey 1999; Danehy and Lock 1995). I assume that Team A trails Team B by one goal. I also assume that the situation—when Team A is contemplating pulling their goalie—is 5-on-5. If Team B scores a goal before Team A, then they will have a two-goal lead, which is assumed to be a lost game for Team A. I address this question within NHL games, mainly because that is where the data is available. There would be little difference at other levels, because scoring rates at all levels of hockey are similar. Within the NHL I assume that all teams are equivalent in their scoring rates. Clearly this is false, but it represents a league-wide average.

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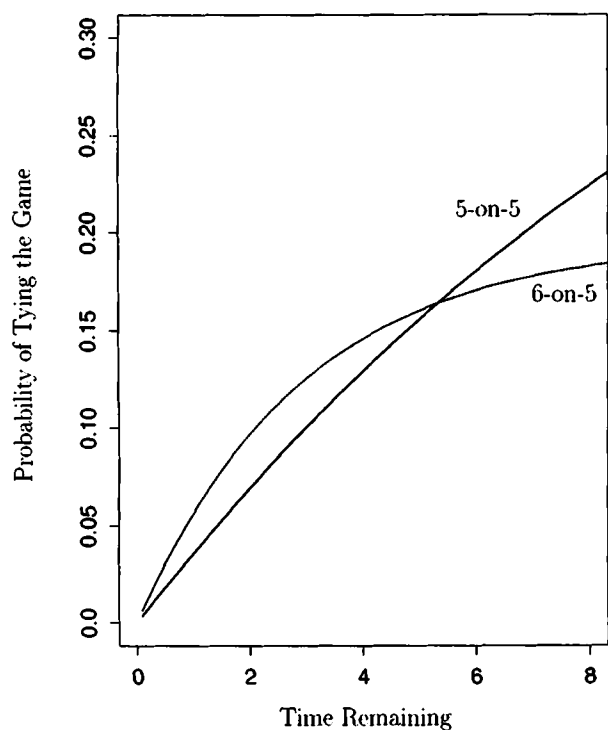


Figure 1. The probability of tying the game if the goalie is pulled (6-on-5) and if the goalie is not pulled (5-on-5) as a function of the time remaining.

To find the probability of Team A tying the game, I need to estimate three parameters. First, if the goalie is not pulled, I need to know the mean time until a goal is scored—when teams are at even strength, 5-on-5. If a team has a 6-on-5 advantage, I need to know the mean time until the team with the extra skater scores a goal and the mean time until they allow an open-net goal. Relevant data are not available for each of these scenarios.

First I estimate the mean time to score in a 5-on-5 scenario. The mean goals per game (which is 60 minutes) for a team is not a great measure for mean goals during 5-on-5. Many goals are scored during a power-play, which means they were scored when one team had more skaters than the other team. In the 1999-2000 NHL season, there were 6,174 total goals scored. Of these, 1,468 were power-play goals and 209 were short-handed goals. The NHL does not keep track of the number of minutes that teams had a man advantage, but they do keep track of the number of advantages. Although most power-plays are two minutes in length, many are shorter. For one reason, when a power-play goal is scored, the power-play is over. It is also possible for a power-play to be five minutes in length. I assume that the average power-play is 1.5 minutes long. Under this scenario, there were 120,831 regular-strength minutes and 13,569 power play minutes (there were 9,046 power-plays). This is a mean difference of 26.86 minutes per goal for an offense during 5-on-5.

I added up the minutes played for each team and the total minutes played by all goalies—there is a difference of 665

minutes. For these 665 minutes with no goalie, there were 182 empty-net goals scored. This is an average rate of one goal every 3.65 minutes. The NHL does not keep track (or at least I couldn't find them) of goals scored for the team that has pulled their goalie. Thus, I need to estimate the rate of scoring when it is 6-on-5. As discussed in the last paragraph there were 13,569 power-play minutes with 1,468 power-play goals scored, for a rate of one goal every 9.24 minutes. Not all power plays are equal. Most are 5-on-4, but some are 5-on-3 or 4-on-3. It is a bigger advantage for the power-play team to have a 5-on-4 advantage than a 6-on-5 advantage (just as it would be better to be 2-on-1 than 11-on-10). A 6-on-5 advantage is then somewhere between one goal every 9.24 minutes and one goal every 26.86 minutes. I think it is closer to the 9.24 minutes. I use the estimate of one goal every 15 minutes. Anderson-Cook and Thornton (1998) addressed the quality of each team's power-play by ranking teams on the average power-play time until a goal. They study the actual power-play time rather than the power-play attempts that the NHL uses. The exponential distribution assumption and the numbers used in this article are consistent with their findings.

Assuming the exponential distribution with a mean time until a goal of λ , the probability of scoring in t minutes is

$$1 - \exp\left(-\frac{t}{\lambda}\right)$$

If Team A has a mean time to score of λ and Team B has a mean time to score of θ , then the probability Team A scores in the next t minutes, before Team B scores, is

$$\left[\frac{\theta}{\theta + \lambda} \right] 1 - \exp\left\{ -t \left(\frac{\lambda + \theta}{\lambda \theta} \right) \right\}$$

For the estimates developed previously, playing 5-on-5 I have $\lambda = \theta = 26.86$ and playing 6-on-5, $\lambda = 15$ and $\theta = 3.65$. Figure 1 shows the probability of Team A tying the game for various times remaining. The break-even point between the two strategies is at the five-minute mark. Thus, anytime with less than five minutes remaining, it is better for Team A to pull their goalie. This does not agree very well with the practiced strategy. In NHL games it is common to see the goalie pulled with about 1.5 minutes left. I have never seen a goalie pulled with more than two minutes remaining (I have seen a goalie pulled in the NHL with about three minutes remaining when a team trailed by two goals). Clearly there are differences in team ability. A high-scoring team would want to pull their goalie later—with less time on the clock—and a low-scoring team would want to pull their goalie sooner. Similar conclusions could be made at all levels of hockey. Differences are in the rate of scoring. If a high school game has a higher rate of scoring when 5-on-5, then the goalie should be pulled later. Would teams pull their goalie with five minutes left after reading this? I do not think they would. Even if they accepted the arguments and assumptions, it would be such a divergence from standard practice that they would be too heavily criticized. The coach is not only trying to win the game but also trying to save his job!

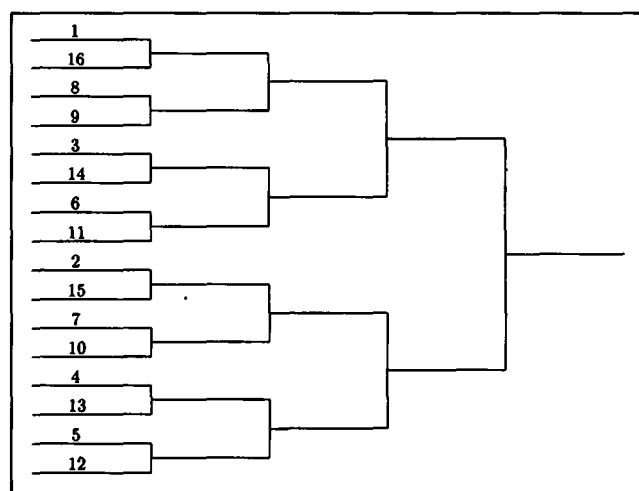


Figure 2. The structure for each region of the NCAA basketball tournament. The numbers refer to the placement for the corresponding seed.

Second Leg: March Madness

The National Collegiate Athletic Association (NCAA) post-season basketball tournament is played throughout the month of March. Famous for the excitement and the upsets, the tournament has become known as "March Madness." It consists of a single elimination 64-team tournament. The 64 teams are split into four sections of 16 teams — each section is called a region. Each team in a region is seeded from 1 to 16, with 1 being the best team. The teams are seeded by an NCAA committee — which I believe does very well.

The schedule of tournament games within each region is shown in Fig. 2. The four winners of the regions meet in a single elimination tournament called the Final Four. There are 15 games in each region, with 3 more in the Final Four, for a total of 63 games.

Part of the popularity of the NCAA March Madness tournament are the office "pools" that are formed. They take many forms, but the general idea is that you have to submit a "bracket" in which you pick the winner of each of the 63 games. In 2000, several Web sites offered \$1,000,000 to anyone who submitted a bracket that gets each of the 63 games correct. How likely is this to happen? Related to this is that it seems every year the media play up how many upsets there are in the tournament—how much madness should we expect? In this leg of this article, I investigate a simple model to predict games in the tournament. I also investigate how many upsets are expected each year.

Breiter and Carlin (1997) and Stern and Mock (1998) both investigated models to predict games between two teams in the NCAA tournament. I construct a model based solely on the seeds for each team — borrowing ideas from each of those articles. First, I steal an idea from Breiter and Carlin. They suggested a model in which if seed i plays seed j then the probability that seed i wins is $j/(i+j)$. They suggested that this can be extended, which is what I do. I give each seed a strength, S_i . Thus, if seed i plays seed j , then the probability that seed i wins is $S_i/(S_i + S_j)$. The strengths for each team are fit using data from each of the 16 (1985–2000) 64-team tournaments (only including games from inside each region). These data are summarized in Table 1.

I impose an additional constraint to the strengths. The strengths cannot be smaller for a team with a better seed.

Table 1 — The Number of Times the Row Seed Has Defeated the Column Seed in the First 16 Years of the NCAA 64-Team Basketball Tournament (only including regional games)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	0	13	5	16	16	4	2	22	33	3	1	10	2	0	0	64
2	10	0	11	2	0	13	28	1	0	14	6	0	0	0	61	0
3	3	7	0	1	1	17	3	1	1	4	11	0	0	51	0	0
4	8	2	1	0	23	2	0	2	1	2	0	7	52	0	0	0
5	3	1	1	15	0	0	0	0	0	0	0	46	6	0	0	0
6	2	4	18	1	0	0	3	0	0	3	45	0	0	8	0	0
7	0	9	1	0	0	1	0	0	0	38	0	0	0	1	1	0
8	7	1	0	3	1	1	1	0	29	0	0	0	1	0	0	0
9	2	0	0	0	1	0	0	35	0	0	0	0	0	0	0	0
10	0	10	2	0	0	2	26	0	0	0	0	0	0	1	2	0
11	1	1	5	0	0	19	1	0	0	0	0	0	0	3	0	0
12	0	0	0	7	18	0	0	0	0	0	0	0	3	0	0	0
13	0	0	0	12	2	0	0	0	0	0	0	1	0	0	0	0
14	0	0	13	0	0	2	0	0	0	0	0	0	0	0	0	0
15	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 2 — The Seed and Strength From the Model of Each Seed

Seed	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Strength	100	43	25	25	21	21	17	17	17	14	10	10	6	5	2	1

Thus a 5-seed must have a strength no lower than the strength for a 6-seed, and so forth. This assumes that when the NCAA selection committee seeds teams they are doing a good job. Although, surely, there are mistakes made, on average I think they do a good job. With this model, one of the strengths must be fixed. Otherwise, multiple sets of strengths produce the same model. For any set of strengths, if each strength is doubled it would result in an identical model. Fixing one strength functions as a form of scaling. An additional problem is that a 16-seed has never won a game in the tournament. This causes a problem — labeled nonidentifiability. To remedy this I fix the strength for both a 16-seed and a 1-seed. To help pick reasonable values, I borrow some information from Stern and Mock. They developed a logistic regression model to predict games from the seed of the two teams. In their article they found that a 16-seed has a probability of about 1/100 of beating a 1-seed. They backed this up by an analysis of the point spreads. Thus, I set $S_1 = 100$ and $S_{16} = 1$. I developed an algorithm to find the maximum likelihood values for the strengths of seeds 2 through 15, subject to the order constraints. The values are given in Table 2.

There is a big separation between a 1-seed and a 2-seed. There is also a fairly large drop between a 2-seed and a 3-seed, but there is little difference between a 3-seed and a 6-seed. Another group with little difference forms in the 7-seed to 10-seed range. A slight drop from there finds the 11-seeds and 12-seeds. There is then a large drop to the 13-seeds and 14-seeds. Yet another drop finds the 15-seeds and the 16-seeds. Table 3 presents the estimated probabilities that an i -seed beats a j -seed.

Using these probabilities and the tournament structure I calculate the probability that each team makes it to the round

of 32, 16, 8, and 4 (these rounds correspond to winning 1, 2, 3, and 4 games, respectively). These probabilities are shown in Table 4. I find the results shown in Table 4 amazing. The probability of advancing in each round is not a monotone function of the seed. For example, the 12-seed has a higher probability of making it to the "Sweet 16" than an 8-seed — even though the 8-seed is better than the 12-seed. This is because the 8-seed would have to beat a 9-seed and a 1-seed, while a 12-seed has to beat a 5-seed and a 4-seed (or maybe a 13-seed!). The 6-seed has a good chance at advancing to the "Great 8" — a better chance than the 4-seed or 5-seed (Hal Stern told me this would happen!). This is solely a function of the bracket structure, not on some strange occurrences in the data. The empirical information backs up this finding — in the 16 years of the 64-team format, an 8-seed and a 9-seed have made the round of 16 only 5 and 2 times, respectively (two 8-seeds made the Final Four in 2000!). The 10-seeds, 11-seeds, and 12-seeds have made 12, 7, and 10 rounds of 16, respectively. This is not because the 10-, 11-, and 12-seeds are better, or that they "have nothing to lose" and "rise to the occasion." It is a simple function of the bracket structure.

Virtually every year the "story" of March Madness is the amazing number of upsets. How many upsets do we expect to see? I simulated 100,000 regional brackets using the probabilities derived in this article. I refer to an upset as anytime a team with a higher seed beats a team with a lower seed. I found a mean of 1.5 upsets in the first round, 1.15 in the second round, .56 in the third round, and .25 in the final round. This is an average of 3.46 upsets per region, which is 13.8 upsets for the whole tournament (not including the Final Four). It is the nature of the tournament that there is not a lot of spread in the quality of the teams — upsets

Table 3 — Each Cell Entry is the Probability That the Row Seed Defeats the Column Seed (probabilities are based on the model)

	1	2	3-4	5-6	7-9	10	11-12	13	14	15	16
1	.50	.71	.80	.83	.85	.88	.91	.94	.95	.98	.99
2	.29	.50	.62	.66	.70	.74	.80	.87	.89	.95	.98
3-4	.20	.38	.50	.54	.60	.64	.71	.81	.83	.93	.96
5-6	.17	.34	.46	.50	.55	.60	.68	.78	.81	.91	.95
7-9	.15	.30	.40	.45	.50	.55	.63	.74	.77	.89	.94
10	.12	.26	.36	.40	.45	.50	.58	.70	.74	.88	.93
11-12	.09	.20	.29	.32	.37	.42	.50	.62	.67	.83	.91
13	.06	.13	.19	.22	.26	.30	.38	.50	.55	.75	.86
14	.05	.11	.17	.19	.23	.26	.33	.45	.50	.71	.83
15	.02	.05	.07	.09	.11	.12	.17	.25	.29	.50	.67
16	.01	.02	.04	.05	.06	.07	.09	.14	.17	.33	.50

Table 4 — The Probability Based on My Model That Each Seed Advances to the Round of 32, 16, 8, and 4

Seed	32	16	8	4
1	.990	.846	.702	.539
2	.956	.701	.474	.196
3	.833	.499	.221	.067
4	.806	.483	.126	.059
5	.677	.351	.082	.035
6	.677	.349	.141	.038
7	.548	.170	.078	.019
8	.500	.077	.035	.013
9	.500	.077	.035	.013
10	.452	.124	.051	.011
11	.323	.113	.028	.005
12	.323	.113	.015	.004
13	.194	.053	.005	.001
14	.167	.040	.006	.001
15	.044	.005	.000	.000
16	.010	.001	.000	.000

are expected — it would be an *upset* if there were not a lot of *upsets*!

How likely is it that a regional bracket goes according to the seeds — that is, that the favorite wins each game? The probability that the favorite wins each of the first round games in a region is .080 (this happened in two regions in 2000). The probability that a region has all the favorites win through the first two rounds is .0145. The probabilities a region has all the favorites win through the third and fourth rounds are .0073 and .0051. The probability that all four regions have the favorites win each of the games is $.0051^4 = 6.9 \times 10^{-10}$. If all the favorites won each of the regional games there would be four 1-seeds in the Final Four. At that point it is probably close to equally likely for each of the games. Thus, the most likely occurrence in the tournament (that all the favorites win) has odds against it of 11.6 billion to 1. Every other combination in the bracket is less likely. These \$1,000,000 offers from the Internet sites may sound enticing, but you're better off saving the time and buying a lottery ticket.

Third Leg: A Nasty Question

The 1998 baseball season was a fantastic one. McGwire and Sosa each broke the single-season home-run record. In *Chance* (Vol. 12, No. 2, Spring 1999) I projected the top 25 home-run hitters for the 1999 season and the number of home runs they would hit. Many popular media sources quoted my predictions of 59 for McGwire and 50 for Sosa. After the 1999 season ended, many people asked me, "Were your predictions correct?" It depends on what your definition of "correct" is (I assume we know what "is" is!). I do not

think they were asking whether I was *exactly* correct ... it would be an incredible statistical mistake to believe that such a goal could be attained. In 1999, McGwire hit 65 and Sosa hit 63. I was off by 6 on McGwire and by 13 on Sosa. Are these far enough from my predicted value that I was *wrong*?

In my model I claimed to model both the quality of the home-run hitters and the number of at-bats. I accounted for minor injuries, but I did not account for the possibility of major injuries. Taking all of these variables as uncertain, the standard deviation in the number of home runs for McGwire was 10.2 and the standard deviation for Sosa was 9.4. So I predicted 59 for McGwire and he hit 65, which was less than one standard deviation away. The *z-score* for McGwire, which is the observed number of home runs minus the predicted, divided by the standard deviation, was $(65 - 59)/10.2 = .59$. I did not do as well for Sosa; he hit 63 home runs although I predicted 50. His *z-score* was $(63 - 50)/9.4 = 1.38$. So, was I right? Maybe my standard errors were so large that the model has too much variability built in. Table 5 presents the predicted home runs, the standard deviation, and the *z-score* for each of the top 25 players.

Three of the players suffered major injuries. Andres Galaraga missed the entire season while undergoing chemotherapy. Barry Bonds batted only 355 times, yet still hit 34 home runs. He missed my projection by 1, but in fairness he would have hit many more. Jay Buhner missed more than half the season, only batting 266 times. I do not consider the results of these three players when addressing the group of players.

From the remaining 22, the worst prediction was for Frank Thomas. He had a miserable year, only hitting 15 home runs. His score was -2.19 . Exactly half of the players had positive *z-scores* (11), but the other half were negative (11). The average of the 22 *z-scores* is .16 with a standard deviation of .96. Theoretically, if I were *right*, the average should be 0 and the standard deviation 1. The predicted distributions for each of the players were not normally distributed — they had longer left tails — but they were fairly close. There were 15 of the 22 *z-scores* between -1 and 1 (68%); for the normal distribution there should be 68%. There was only one larger, in magnitude, than 2 (4.5%); for the normal distribution this is 5%. This matches very well to my predictions.

Although none of the players hit exactly what I predicted for them, I still believe I was correct. Is it possible that I was correct but someone else was more correct? I do not know of anyone else that predicted home runs for these players, so I use a different competitor — Mr. MLE. I use the number of home runs hit in 1998 as the competition. So Mr. MLE predicts that each player will hit the same number in 1999 as they did in 1998. The 1998 home runs are also listed in Table 5.

Mr. MLE beat my predictions on 11 of the 22 players, but mine were better on 10, and they were equal on another. The mean absolute error in prediction using the 1998 total is 6.64, whereas it was 6.81 for my projections. The standard deviation of the errors is 7.65 for each projection method. The mean of the errors is 2.36 for Mr. MLE and -1.45 for my predictions. If *correct* is measured by doing better than the simple method of Mr. MLE, then I was not correct. It is not surprising that the methods were so close

Table 5 — The Predicted Top 25 Home-Run List for 1999

Rank	Player	My prediction	St. dev.	Actual	Z-score	1998
1	Mark McGwire	59	10.2	65	.58	70
2	Ken Griffey	51	9.4	48	-.31	56
3	Sammy Sosa	50	9.4	63	1.38	66
4	Juan Gonzalez	44	8.8	39	-.57	45
5	Vinny Castilla	39	8.3	33	-.72	46
6	Albert Belle	39	8.2	37	-.24	49
7	Jose Canseco	39	8.5	34	-.59	46
8	Andres Galarraga	39	9.3	0	-4.19	44
9	Rafael Palmeiro	37	8.1	47	1.23	43
10	Greg Vaughn	37	8.1	45	.98	50
11	Mo Vaughn	37	8.0	33	-.50	40
12	Manny Ramirez	35	7.9	44	1.14	45
13	Barry Bonds	35	7.8	34	-.13	37
14	Jeff Bagwell	34	7.8	42	1.03	34
15	Jim Thome	35	8.0	33	-.25	30
16	Alex Rodriguez	34	7.8	42	1.03	42
17	Vladimir Guerrero	34	8.6	42	.93	38
18	Jay Buhner	34	8.2	14	-2.44	15
19	Mike Piazza	34	7.7	40	.77	32
20	Nomar Garciaparra	33	8.8	27	-.68	35
21	Larry Walker	33	7.9	37	.50	23
22	Carlos Delgado	33	7.6	44	1.44	38
23	Tino Martinez	32	7.5	28	-.53	28
24	Henry Rodriguez	32	7.6	26	-.79	31
25	Frank Thomas	31	7.3	15	-2.19	29

Note: The "My prediction" and "St. dev." columns are the mean and standard deviation of my predictive distribution. The "Actual" column refers to the number of home runs in 1999. The "Z-score" column is the standardized scores based on my predicted mean and standard deviations. The "1998" column reports the number of home runs each player hit in 1998.

— my projection is essentially an adjustment to the 1998 totals (using league-wide averages and the 1997 and 1996 totals). I am disappointed that my projections didn't beat Mr. MLE — I still think it will do better in the future and better for the whole league. Was I correct? Yes ... well, okay, at least I wasn't wrong.

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References and Further Reading

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Where are the Data?

The NCAA 64 data from Table 1 and the data used for the home-run article are available at my Web site, stat.tamu.edu/~berry.

nhl.com : This is the official site of the NHL. It has data available for the current season, but little on past seasons.

cbs.sportsline.com/u/madness/2000/history/yearbyyear: This site, affiliated with CBS Sports, provides information for each year of the basketball championship. Although, not in a great format, each game's results and scores can be found.